

# Nathanael Hadgu

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## Education

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North Dakota State University | Fargo, ND **Bachelor of Science in Software Engineering** | Expected: Dec 2027

## Technical Skills

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**Programming & Data:** Java, Python, JavaScript, SQL

**Developer Tools & Environments:** Git, GitHub, Linux/Bash, VS Code

**Hardware & Prototyping:** Fusion 360, Additive Manufacturing (3D Printing), Breadboarding

## Technical Projects

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**Expression Parser and Evaluator** | *Java, Custom Data Structures*

- Built a command-line arithmetic expression evaluator implementing Dijkstra's Shunting-Yard algorithm to parse infix notation into Reverse Polish Notation (RPN).
- Designed custom generic Stack and Queue data structures to manage operator precedence, parentheses matching, and token evaluation from scratch without external libraries.
- Handled edge-case syntax validation, unbalanced brackets, and divide-by-zero exceptions with robust error-handling routines.

**Space Invaders Arcade Clone** | *Java, OOP*

- Developed a 2D desktop arcade game utilizing object-oriented design patterns to manage entity lifecycles, player states, and wave progression.
- Implemented axis-aligned bounding box (AABB) collision detection and custom game-loop timing for synchronized projectile rendering and enemy movement.
- Encapsulated game logic across distinct classes (Player, Enemy, Projectile) to decouple UI rendering from physics and state management.

## Experience

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Digital Fabrication Lab Assistant | NDSU Libraries, Fargo, ND

**February 2025 - Current**

- Supported digital fabrication operations, assisting students and faculty in preparing CAD models and running jobs across rapid prototyping equipment.
- Calibrated, maintained, and operated 3D printers and laser fabrication machinery, troubleshooting mechanical errors and slicing configurations to ensure high print fidelity.
- Managed material intake, queue prioritization, and project fulfillment tracking for cross-departmental capstone and research commissions.

## Leadership Experience

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**Recruitment Chair** | *NDSU Bison Robotics*

**March 2025 – Current**

- Drove recruitment strategy across campus, growing team membership by 15% in one semester through data-informed outreach and targeting technical student organizations.
- Delivered technical program overviews and pitch presentations to audiences of 200+ prospective engineering students.